

Luke Bielawski

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EDUCATION

University of Illinois Urbana-Champaign

Bachelor of Science in Mathematics, Bachelor of Science in Physics
Minor in Computer Science

Champaign-Urbana, IL

Expected May 2027

- Cumulative GPA: 3.98 / 4.0
- Honors & Awards: Chancellor's Scholar, LAS James Scholar, Dean's List
- Relevant Coursework: Linear Algebra, Real Analysis, Smooth Manifolds I & II, Probability Theory (UG, G), Measure Theory, Classical Mechanics I & II, Electrodynamics I & II, Quantum Mechanics, Data Structures

RESEARCH

University of Maryland

(NSF REU) TREND: Collective Behavior

College Park, MD

June 2025 - August 2025

- Developed a kinetic equation and proved existence of solutions using a variety of tools from analysis
- Explored flocking and stochastic adjustments, analytically and numerically solving SDEs
- Collaborated with faculty and peers to explore computational methods for efficient calculation
- Publication of work in progress

Illinois Institute of Technology

(NSF REU) SURE: Speedier Simulations

Chicago, IL

May 2024 - August 2024

- Investigated low discrepancy sequences and their application to kernel density estimation
- Researched and implemented quasi-Monte Carlo density estimation with conditioning into QMCPy
- Presented research biweekly for other students and faculty utilizing LaTeX and Beamer

University of Illinois Urbana-Champaign

Illinois Mathematics Lab: Linear Poisson Structures and their Cohomology

Champaign-Urbana, IL

Jan 2024 - May 2024

- Calculated Cohomology groups for various linear Poisson structures via power series
- Wrote Python code to automatically calculate Cohomology classes with finite degree power series
- Exhibited findings at Undergraduate Research Symposium and IML Poster Session

WORK EXPERIENCE

University of Illinois Urbana-Champaign

Undergraduate Course Assistant

Champaign-Urbana, IL

August 2025 - Present

- Running biweekly discussion, teaching course material and helping with problem solving
- Holding weekly office hours; answering student questions; proctoring and grading exams

Varsity Tutors

Remote Tutor for ACT, SAT, Competition Math

Arlington Heights, IL

June 2023 - August 2023

- Provided individual instruction to 6 students on a weekly basis over a two-month span
- Measured student progress through regular testing, adapting instruction to match individual needs
- Taught content as well as test-taking techniques to best prepare students for their academic goals

LEADERSHIP EXPERIENCE

Chi Psi Fraternity

Scholarship Chair

Champaign-Urbana, IL

Dec 2023 - May 2024

- Facilitated study hours for the brothers to promote productive and consistent study habits
- Offered support and strategizing with academically struggling brothers to best improve
- Coordinated with alumni and brothers on internal scholarships and academic purchases

ADDITIONAL INFORMATION

Technical Skills: Python, C++, MATLAB, Mathematica, Microsoft Excel, Italian

Academic Interests: Differential Geometry, Probability Theory, Mathematical Physics

Other Interests: Putnam Math Competition, Quantitative Finance, Flute and Music